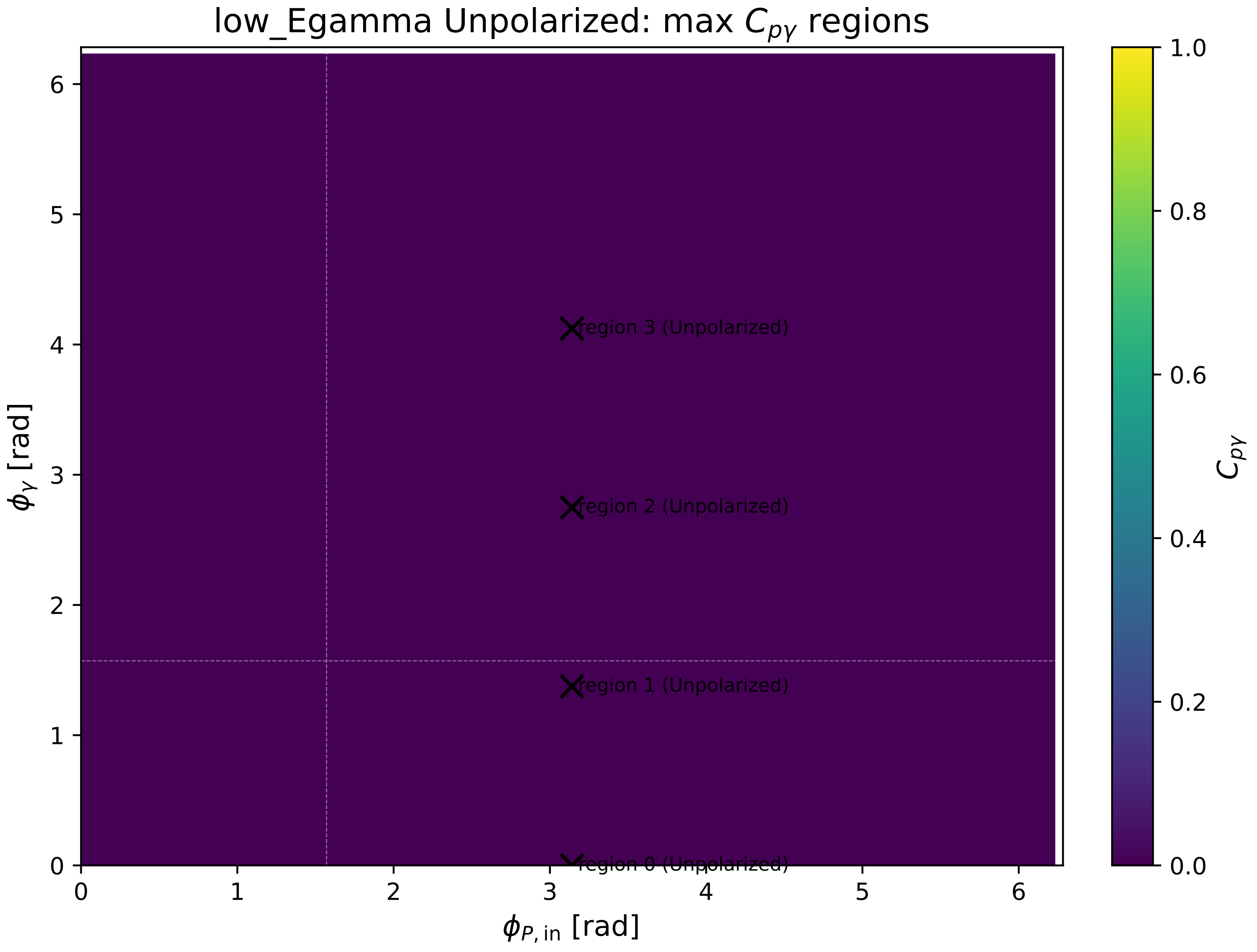
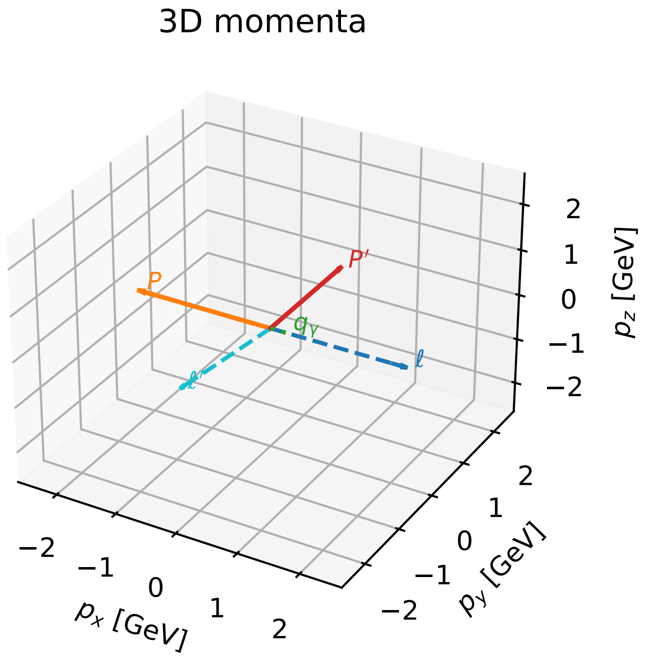


low_Egamma Unpolarized: max $C_{p\gamma}$ regions
region 3 (Unpolarized)
region 2 (Unpolarized)
region 1 (Unpolarized)
region 0 (Unpolarized)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{p\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Unpolarized characteristic max $C_{p\gamma}$ configuration

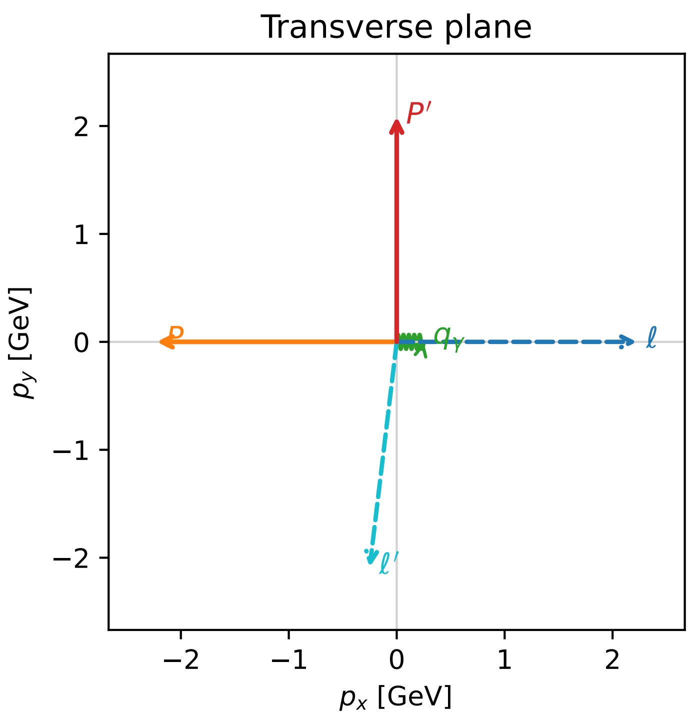


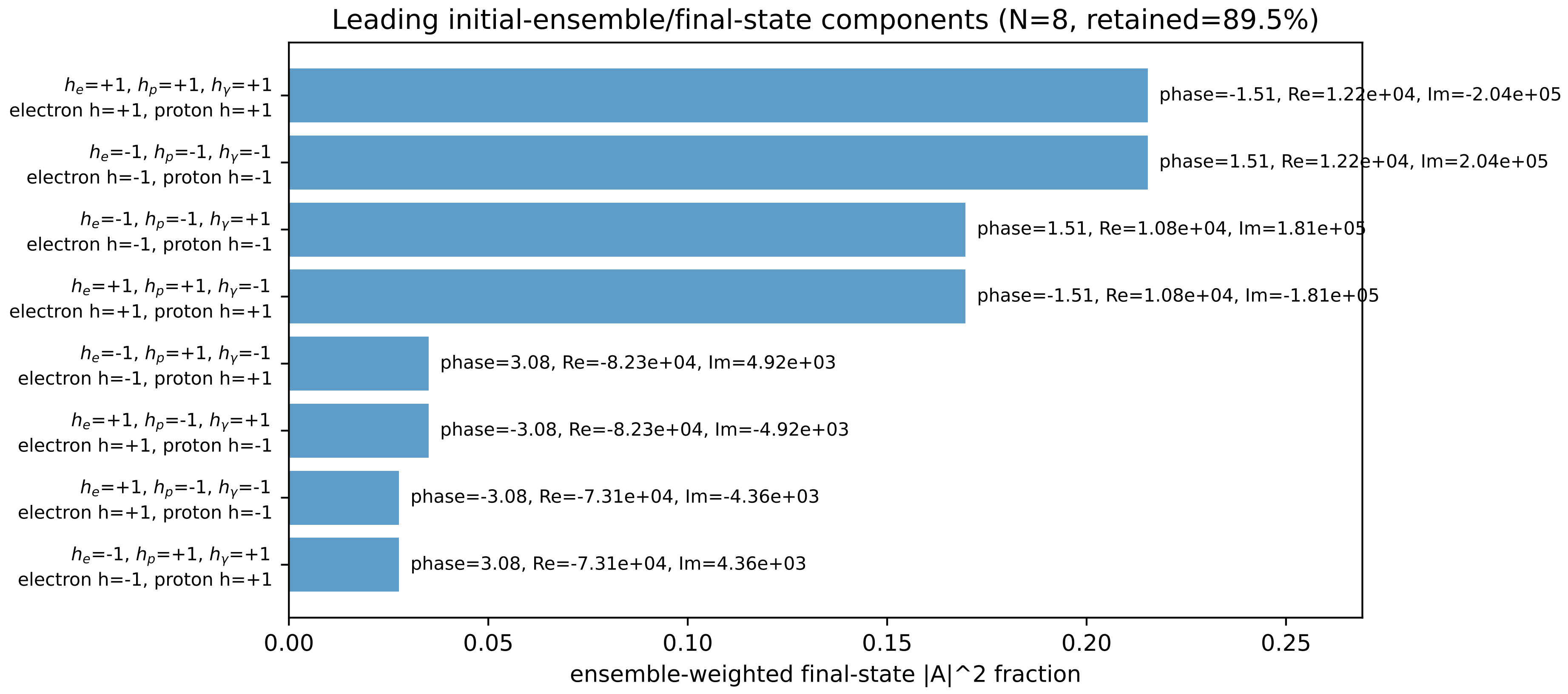

```
c_p_gamma_low_egamma_unpolarized_region_0  C_{p\gamma}=0
region: low_Egamma_unpolarized_region_0  final-pair delta_xy=1.5708 rad
outgoing-state purity: 0.368276
kinematic point: high_s_high_theta_in_low_Egamma
```

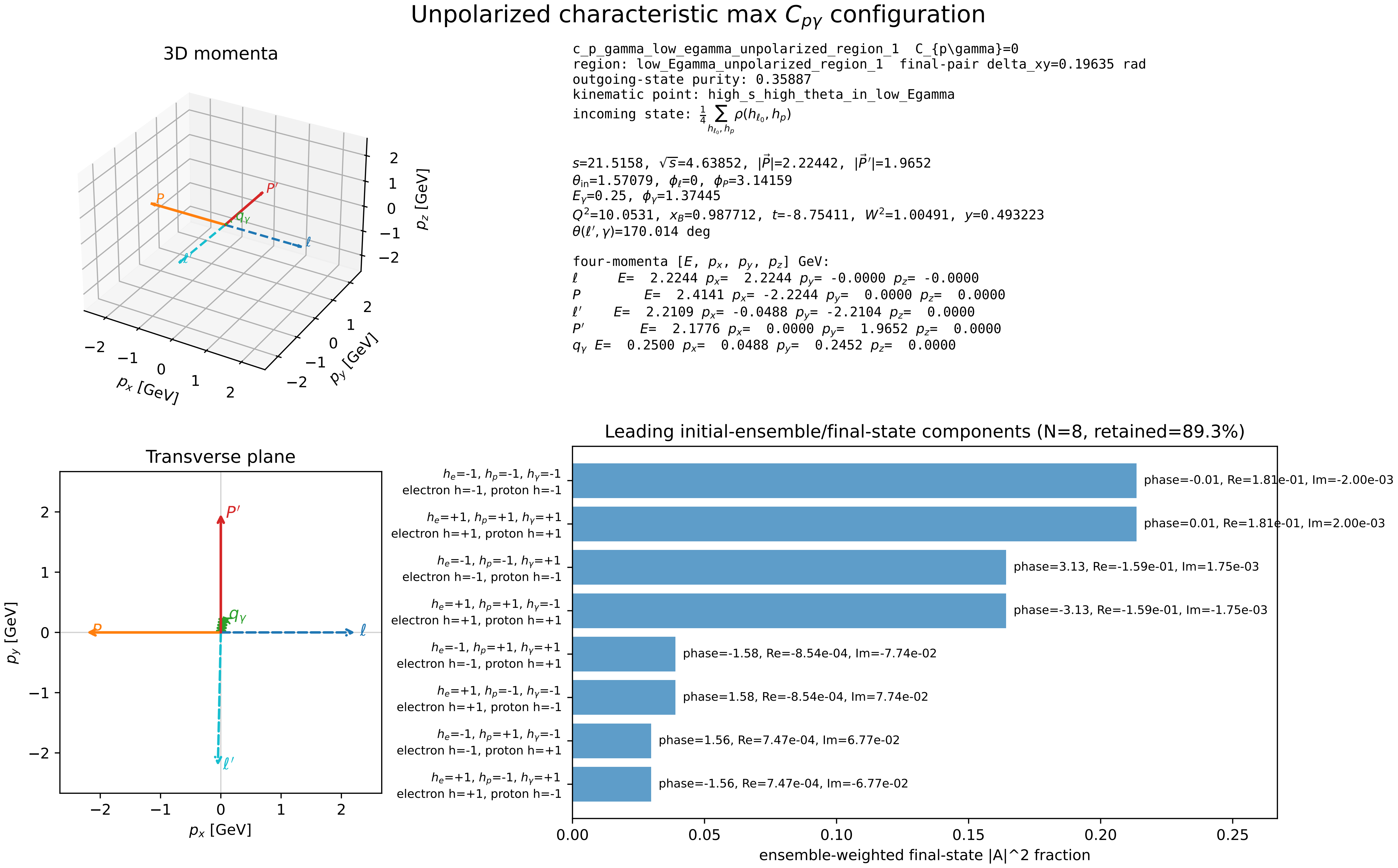
incoming state: $\frac{1}{4}\sum_{h_{\ell_0},h_p}\rho(h_{\ell_0},h_p)$

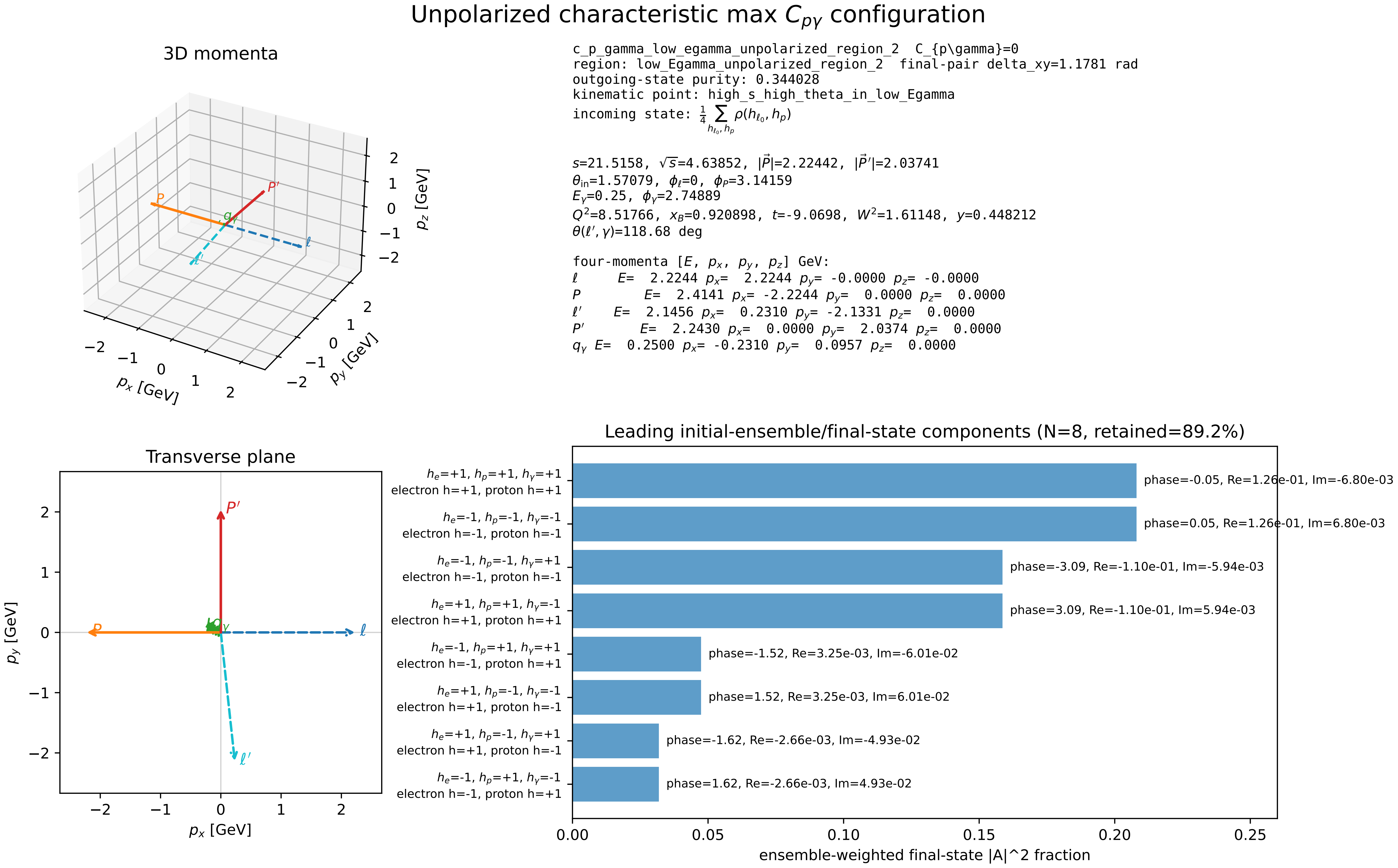
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.08621
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=0.25, ϕ_γ=0
Q^2=10.4598, x_B=0.901436, t=-9.28425, W^2=2.02353, y=0.562294
$\theta(\ell',\gamma)$=96.8334 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 2.1011 p_x= -0.2500 p_y= -2.0862 p_z= 0.0000
P' E= 2.2874 p_x= 0.0000 p_y= 2.0862 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2500 p_y= 0.0000 p_z= 0.0000



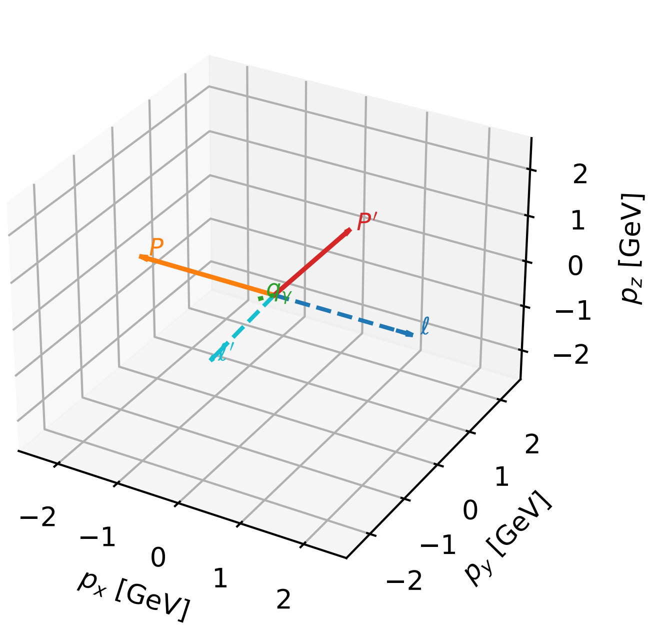



Unpolarized characteristic max $C_{p\gamma}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
c_p_gamma_low_egamma_unpolarized_region_1 C_{p\gamma}=0
region: low_Egamma_unpolarized_region_1 final-pair delta_xy=0.19635 rad
outgoing-state purity: 0.35887
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{4}\sum_{h_{\ell_0},h_p}\rho(h_{\ell_0},h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.9652
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=0.25, ϕ_γ=1.37445
Q^2=10.0531, x_B=0.987712, t=-8.75411, W^2=1.00491, y=0.493223
$\theta(\ell',\gamma)$=170.014 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 2.2109 p_x= -0.0488 p_y= -2.2104 p_z= 0.0000
P' E= 2.1776 p_x= 0.0000 p_y= 1.9652 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.0488 p_y= 0.2452 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=8, retained=89.3%)
h_e=-1, h_p=-1, h_γ=-1 electron h=-1, proton h=-1
phase=-0.01, Re=1.81e-01, Im=-2.00e-03
h_e=+1, h_p=+1, h_γ=+1 electron h=+1, proton h=+1
phase=0.01, Re=1.81e-01, Im=2.00e-03
h_e=-1, h_p=-1, h_γ=+1 electron h=-1, proton h=-1
phase=3.13, Re=-1.59e-01, Im=1.75e-03
h_e=+1, h_p=+1, h_γ=-1 electron h=+1, proton h=+1
phase=-3.13, Re=-1.59e-01, Im=-1.75e-03
h_e=-1, h_p=+1, h_γ=+1 electron h=-1, proton h=+1
phase=-1.58, Re=-8.54e-04, Im=-7.74e-02
h_e=+1, h_p=-1, h_γ=-1 electron h=+1, proton h=-1
phase=1.58, Re=-8.54e-04, Im=7.74e-02
h_e=-1, h_p=+1, h_γ=-1 electron h=-1, proton h=+1
phase=1.56, Re=7.47e-04, Im=6.77e-02
h_e=+1, h_p=-1, h_γ=+1 electron h=+1, proton h=-1
phase=-1.56, Re=7.47e-04, Im=-6.77e-02
ensemble-weighted final-state |A|^2 fraction



Unpolarized characteristic max $C_{p\gamma}$ configuration
3D momenta
P
P'
q_γ
ℓ
ℓ'
p_x [GeV]
p_y [GeV]
p_z [GeV]
c_p_gamma_low_egamma_unpolarized_region_2 C_{p\gamma}=0
region: low_Egamma_unpolarized_region_2 final-pair delta_xy=1.1781 rad
outgoing-state purity: 0.344028
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{4}\sum_{h_{\ell_0},h_p}\rho(h_{\ell_0},h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.03741
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=0.25, ϕ_γ=2.74889
Q^2=8.51766, x_B=0.920898, t=-9.0698, W^2=1.61148, y=0.448212
$\theta(\ell',\gamma)$=118.68 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 2.1456 p_x= 0.2310 p_y= -2.1331 p_z= 0.0000
P' E= 2.2430 p_x= 0.0000 p_y= 2.0374 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.2310 p_y= 0.0957 p_z= 0.0000
Transverse plane
Leading initial-ensemble/final-state components (N=8, retained=89.2%)
h_e=+1, h_p=+1, h_γ=+1
electron h=+1, proton h=+1
phase=-0.05, Re=1.26e-01, Im=-6.80e-03
h_e=-1, h_p=-1, h_γ=-1
electron h=-1, proton h=-1
phase=0.05, Re=1.26e-01, Im=6.80e-03
h_e=-1, h_p=-1, h_γ=+1
electron h=-1, proton h=-1
phase=-3.09, Re=-1.10e-01, Im=-5.94e-03
h_e=+1, h_p=+1, h_γ=-1
electron h=+1, proton h=+1
phase=3.09, Re=-1.10e-01, Im=5.94e-03
h_e=-1, h_p=+1, h_γ=+1
electron h=-1, proton h=+1
phase=-1.52, Re=3.25e-03, Im=-6.01e-02
h_e=+1, h_p=-1, h_γ=-1
electron h=+1, proton h=-1
phase=1.52, Re=3.25e-03, Im=6.01e-02
h_e=+1, h_p=-1, h_γ=+1
electron h=+1, proton h=-1
phase=-1.62, Re=-2.66e-03, Im=-4.93e-02
h_e=-1, h_p=+1, h_γ=-1
electron h=-1, proton h=+1
phase=1.62, Re=-2.66e-03, Im=4.93e-02
ensemble-weighted final-state |A|^2 fraction

Unpolarized characteristic max $C_{p\gamma}$ configuration




```
c_p_gamma_low_egamma_unpolarized_region_3  C_{p\gamma}=0
region: low_Egamma_unpolarized_region_3  final-pair delta_xy=2.55254 rad
outgoing-state purity: 0.348285
kinematic point: high_s_high_theta_in_low_Egamma
```

incoming state: $\frac{1}{4}\sum_{h_{\ell_0},h_p}\rho(h_{\ell_0},h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.19996
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=0.25, ϕ_γ=4.12334
Q^2=8.26611, x_B=0.796618, t=-9.78736, W^2=2.99024, y=0.502835
$\theta(\ell',\gamma)$=37.7383 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.9969 p_x= 0.1389 p_y= -1.9921 p_z= 0.0000
P' E= 2.3916 p_x= 0.0000 p_y= 2.2000 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1389 p_y= -0.2079 p_z= 0.0000